


Problem 1

Y_i = final score; X_i = percent attend:

$$(a) E[Y_i | X_i] = \beta_0 + \beta_1 X_i \Rightarrow \frac{\partial E[Y_i | X_i]}{\partial X_i} = \beta_1, \quad \hat{\beta}_1 = 0.09672$$

(b) Standard test of significance: $H_0: \beta_1 = 0$ vs $H_a: \beta_1 \neq 0$

$$T = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \sim t(680 - 2) = t(678)$$

$$T^{obs} = \frac{0.09672}{0.02627} \approx 3.68 \quad \text{compare this to } t_{0.025/2}^{678} = t_{0.025}^{678} = 1.96$$

Since $|T^{obs}| = 3.68 > 1.96 \rightarrow \text{reject } H_0$

$$(c) \Delta Y = \beta_1 \Delta X = 20 \beta_1 \quad H_0: \Delta Y = 20 \beta_1 = 2 \Leftrightarrow \beta_1 = \frac{2}{20} = 0.1$$

vs

$$H_a: \Delta Y = 20 \beta_1 < 2 \Leftrightarrow \beta_1 < 0.1$$

$$T = \frac{\hat{\beta}_1 - 0.1}{SE(\hat{\beta}_1)} \sim t_{(678)}$$

$$T_{obs} = \frac{0.09672 - 0.1}{0.02627} \approx -0.1248$$

Left-tailed critical value $t_{0.01}^{0.78} \approx -2.326 \rightarrow$ We can't reject H_0 .

$$(c) \quad \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} \Rightarrow \bar{y} = 64.72794$$

$$\text{Formula } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$(e) \quad \widehat{Var}(y|x) = \hat{\sigma}^2 = 11.67^2$$

$$Var(y|x) = \sigma^2$$

$$(f) \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 100 = 66.4968$$

$$(g) 1) E(\hat{\beta}_0 + 100\hat{\beta}_1 | X's) = E(\hat{\beta}_0 | X's) + 100 E(\hat{\beta}_1 | X's) \\ = \beta_0 + 100\beta_1$$

$$2) \text{Var}(\hat{\beta}_0 + 100\hat{\beta}_1 | X's) = \text{Var}(\hat{\beta}_0 | X's) + 100^2 \text{Var}(\hat{\beta}_1 | X's) + 2(100) \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's)$$

$$\text{Estimates: } 1) \hat{\beta}_0 + 100\hat{\beta}_1 = 66.4968$$

$$2) 4.80 + 10,000(0.00069) + 200(-0.05637) \approx \underline{0.432}$$

$$(h) T = \frac{\hat{Y} - (\beta_0 + \beta_1 100)}{SE(\hat{Y})} \Rightarrow CI: \hat{Y} \pm t_{0.025}^{678} SE(\hat{Y})$$

$$\hat{Y} = 66.49$$

$$CI = [65.2066, 67.705]$$

$$SE(\hat{Y}) = \sqrt{0.432}$$

(i) Bias:
$$\delta_2 = \frac{\sum_i (x_i - \bar{x})(z_i - \bar{z})}{s_{xx}}$$

z_i = homework

δ_2 : Ceteris Paribus effect of homework on final score, should be positive.

$\sum_i (x_i - \bar{x})(z_i - \bar{z})$; if positive, bias positive, not including homework biases upwards the estimate of the effect of attendance.

(j) # Class Taken

Examples

- GPA at the year
- Mental Health of student
- Grade in statistics

Problem 2

$$(a) \quad \min_c \sum_i (X_i - c)^2$$

$$\text{F.O.C:} \quad \sum_i -2(X_i - c^*) = 0$$

$$\Leftrightarrow \sum_i (X_i - c^*) = 0$$

$$\Leftrightarrow \sum_i X_i = \sum_i c^* = n c^* \Leftrightarrow c^* = \frac{1}{n} \sum_i X_i$$

$$\text{OR } c^* = \bar{X}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \frac{2}{3} + a^2 - a^2 \\ &= 2/3 \end{aligned}$$

$$(b) \quad \bar{X}_n \xrightarrow{p} E(X) = a \quad \text{by WLLN} \quad (\bar{X}_n \text{ is consistent, } \bar{X}_n \text{ is the appropriate estimator})$$

$$\lim_{n \rightarrow \infty} E[(\bar{X}_n - a)^2] = \lim_{n \rightarrow \infty} (\text{Var}(\bar{X}_n) + (E(\bar{X}_n) - a)^2) \quad E(\bar{X}_n) = a$$

$$= 0 + (a - a)^2 = 0$$

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \frac{\text{Var}(X)}{n} \\ &= \frac{2/3}{n} \end{aligned}$$

$$\Rightarrow \text{convergence in } L_2$$

(c) Use the CLT

$$\frac{\bar{X}_n - a}{\sqrt{\frac{2}{3n}}} \overset{a}{\sim} N(0, 1) \Rightarrow \bar{X}_n \overset{a}{\sim} N(a, \frac{2}{3n})$$

$$\begin{aligned} \text{(d)} \quad \Pr(|\bar{X}_n - a| \leq 0.05) &= \Pr\left(\frac{|\bar{X}_n - a|}{\sqrt{\frac{2}{3n}}} \leq \frac{0.05}{\sqrt{\frac{2}{3n}}}\right) \\ &= \Pr(|N(0, 1)| \leq \frac{0.05}{\sqrt{\frac{2}{3n}}}) \end{aligned}$$

When $n = 3,650$

$$\Pr(|N(0, 1)| \leq 3.6996)$$

$$= \Pr(-3.6996 \leq N(0, 1) \leq 3.6996)$$

$$= \text{CDF}(3.6996) - \text{CDF}(-3.6996) \approx 1$$

$$(e) \quad \hat{a} = \bar{x} = \frac{29,200}{3650} = 8$$

$$f(x; 8) = \frac{1}{4} (2 - 8 + x) I_{[6, 8]}(x) + \frac{1}{4} (2 + 8 - x) I_{(8, 10]}(x)$$

$$\Pr(X > 21) = 1 - \Pr(X \leq 21) = 0$$



$$\Pr(X \leq 21) = \int_{-\infty}^{21} f(x; 8) dx = 1$$